


COMPUTER SECURITY
AND
DIGITAL TRANSFORMATION
COURSEWORK
HANDBOOK
AY: 2025-26

2

Policies and Procedures

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1 Final Output

2.1.1 Unless specified otherwise in the topic description, a complete final submission must include the **report PDF**, the required editable source files, and any additional artefacts required by the project description, **including code, data, spreadsheets, notebooks, appendices, or repository links where applicable.**

2.1.2 You are **encouraged to use your Microsoft 365** account for group collaboration and the **free Zotero plugin for the bibliography.**

2.1.3 You can use the free Overleaf account or any other installation you prefer for LaTeX. For word counting, you can use TeXcount¹ which is integrated into Overleaf.²

2.1.4 Unless specified otherwise in the topic description, the maximum length for the report is summarised in Table 2.1. This count excludes tables, figures, captions, acknowledgements, references, appendixes, and executive summary. **An executive summary is mandatory**, and must be limited to 400 words.

0 IN TEAMS

¹ <https://app.uio.no/ifi/texcoun t/> (on 28 Feb 2025).

² <https://www.overleaf.com/blog/ word-count-2015-09-15> (on 28 Feb 2025).

Table 2.1: Report maximum length: the three cases. A +10% grace is automatically granted for all cases. This count excludes tables, figures, captions, acknowledgements, references, appendixes, and executive summary. The **mandatory executive summary must be limited to 400 words.**

³ <https://creativecommons.org/licenses/by/4.0/> (on 1 April 2026).

⁴ <https://creativecommons.org/licenses/by/4.0/> (on 1 April 2026).

⁵ <https://opensource.org/licenses/mit> (on 1 April 2026).

	Word limit
Computer Security only	3 000
Digital Transformation only	5 000
Joint Computer Security and Digital Transformation	7 000

2.1.5 The final technical report must be written in **English** and released under the **CC-BY 4.0 licence**.³

2.1.6 If the project requires the production of other artefacts, e.g., spreadsheets, they will have to be shared in the original format, e.g., Excel file, under the **CC-BY 4.0 licence**.⁴

2.1.7 If the project requires code production, the source code should be shared with the **MIT licence**.⁵

2.1.8 These licensing requirements apply only to material produced by the group. Any third-party material included in the submission must be used in compliance with its own licence or terms of use and must be clearly attributed.

2.1.9 The final mark is based on Deliverable 1 and Deliverable 3 as submitted, and the history of interaction between the group and the lecturer. → **DOCUMENT-BASED**

2.1.10 If the written material does not provide sufficient evidence to determine the mark, the lecturer may require an oral defence. In that case, the oral defence becomes a necessary component of the evaluation. This will involve an in-person discussion of the submitted material. The group will be allowed five minutes to summarise their submission before any necessary aspects are examined in detail. (ON THINGS NOT UNDERSTOOD)

← "DEFENCE"

2.1.11 Each group can submit a supplementary document detailing the specific contributions of each member. If submitted, this document will determine whether individual group members should receive different marks. If no such document is provided, all group members will receive the same mark, and they will accept collective responsibility for all parts of the coursework. → **ORAL # ON #**

2.1.12 Any material violation of the policies set out in this rulebook is sufficient grounds for assigning a fail mark to the coursework. A material violation is any breach that, in the lecturer's judgement, compromises academic integrity, authorship transparency, legal or licensing compliance, fairness of assessment, or the possibility of properly evaluating the submitted work. Such violations include, but are not limited to, undisclosed substantive use of generative AI where disclosure is required, plagiarism, fabrication or falsification of results, concealment of conflicts of interest, submission of materially incomplete deliverables, or failure to provide required

source code or source files where these are mandatory.

2 Submissions Deadline and Reassessment

- 2.2.1 The final output — **fully completed and meeting all requirements, as outlined in Section 1** — must be submitted via Microsoft Teams by the deadlines specified in Section 2. Submissions deferred to a subsequent academic year must comply with the handbook in force for that academic year.
- 2.2.2 Section 2 also provides the expected timeline for the potential scheduling of the group oral defence and the release of final grades.
- 2.2.3 It is the student's responsibility to plan their activities by the deadlines imposed by the university for graduation⁶ or by the exchange programme they are enrolled in (e.g., Erasmus, ...).
- 2.2.4 Students seeking reassessment must complete a different project and submit it in a subsequent submission cycle. Reassessment may be undertaken in a newly formed group, subject to project availability and approval.

⁶ See <https://www.unibs.it/en/education/studying/graduating/engineering> (on 1 April 2026).

Submission Deadline	Possible Oral Defence [‡]	Expected Grade Release [‡]
08 June 2026 09:00 am CEST	17 June 2026 [‡]	19 June 2026 [‡]
22 June 2026 09:00 am CEST	2 July 2026 [‡]	7 July 2026 [‡]
31 August 2026 09:00 am CEST	8 September 2026 [‡]	11 September 2026 [‡]
First Day of the Winter Exam Session 7 January 2027 [‡] 09:00 am CET	TBD	TBD
Third Monday of the Winter Exam Session 25 January 2027 [‡] 09:00 am CET	TBD	TBD
First Day of the Easter Exam Session 22 March 2027 [‡] 09:00 am CET	TBD	TBD

Table 2.2: Coursework submission deadlines. Unforeseen events could modify the dates marked with [‡].

3 Coursework Activity Phases

2.3.1 This section outlines the structured process for completing the coursework. The work is divided into four sequential phases.

2.3.2 The four sequential stages are:

- Group formation and project allocation; 2-3 (1-4 POSSIBLE)
- Preparation of a high-level programmatic document (Deliverable 1, **this is mandatory**);
- Execution of the project and preparation of an outline of results, which can be **optionally** submitted for discussion with the lecturer (Deliverable 2);
- **Final submission** (Deliverable 3, **no revisions will be allowed**).

2.3.3 Table 2.3 summarises the ethos of the three requested deliverables and their word limits, which are described in the following.

2.3.4 The description below **refers extensively to Federico Cerutti and Timothy J. Norman. Notes on Researching and Writing. 2025**, referenced as [Notes Writing]. This document is present in the online material of the module.⁷

⁷ Access it via <https://elearning.unibs.it/> (on 1 April 2026).

2.3.5 Some coursework projects are co-supervised with other members of staff. In such cases, the guidelines for completing the coursework may differ.

	Deliverable	Activities	Word Limit
1	High-Level Programmatic Document	Define subject, justify importance, outline communication strategy. Share via Teams, discuss with lecturer, obtain approval. Provide changelog if revised.	1 000 (+ 10% grace)
2	Project Execution and Outline of Results Optional deliverable submission	Conduct research, identify thesis and arguments, prepare a detailed outline. Submit via Teams, wait for feedback, provide changelog if revised.	Half of word limits in Table 2.1 (with grace)
3	Final Write-Up	Expand outline into full draft, refine for clarity and coherence, produce final report.	See Table 2.1

Table 2.3: Summary of Deliverables and Activities

Group Formation and Project Allocation

Group Formation and Project Allocation

- 2.3.6 Students must form groups and bid for two projects: a primary choice and an alternative. Bidding should be done via email, including:
- The full names of all group members (who must be CCed in the email);
 - A brief justification (up to two lines) for each project choice, if applicable. Justifications may include relevant expertise, such as prior experience with LLMs or participation in the Computer Security module;
 - A comprehensive and transparent disclosure of any actual, potential, or perceived conflicts of interest. Conflicts of interest may arise from, but are not limited to:
 - Overlaps in content, data, or methodologies with other academic work, including coursework, research projects, theses, or internships;
 - Employment, consultancy, or professional activities that relate to or may benefit from the project outcomes;
 - Prior or ongoing collaborations with individuals, organisations, or institutions that may have an interest in the project results.

Declaring a conflict of interest does not automatically disqualify a group from being assigned a particular project. Instead, where a conflict is disclosed, a meeting will be arranged with all relevant stakeholders to determine an appropriate course of action. This may involve implementing additional oversight, modifying project parameters, or agreeing on specific guidelines to ensure academic integrity, fairness, and compliance with institutional policies.

- 2.3.7 Groups must wait for the lecturer to confirm their allocation before proceeding. Once confirmed, the lecturer will set up a dedicated Microsoft Teams channel, which will serve as the main communication platform unless the project involves co-supervision.

- 2.3.8 The expected size of groups should be either 2 or 3. Under exceptional circumstances, the lecturer reserves the right to allow single students undertaking a project as well as groups of at most size 4.

Deliverable 1: High-Level Programmatic Document

Deliverable 1

- 2.3.9 The first deliverable is a high-level programmatic document.⁸

⁸ See [Notes Writing, Ch. 2] for details.

- 2.3.10 This document must address the following key questions:

- What is the specific subject of study?
- Why is this subject important? Who is the intended audience? What impact should the project have on their understanding?
- How will the project achieve its communication objectives?

- 2.3.11 It must be **shared via Microsoft Teams**.
- 2.3.12 **Approval is required before moving forward**. Discussions typically occur asynchronously on Microsoft Teams but may also be raised during lectures.
- 2.3.13 **If revisions are requested, a full changelog must be provided**. Each group will be allowed at **most two revisions** (excluding the original submission). If, after these revisions, the lecturer determines that the deliverable still does not meet the required standard, the **group will be required to abandon the current project and start over with a new one**.
- 2.3.14 **The final output should remain materially consistent with the last approved version of Deliverable 1**. Any substantive change in scope, research question, methodology, or intended output **must be explicitly justified, either in Deliverable 2 or, if Deliverable 2 is not submitted, in Deliverable 3**.

Deliverable 2 (optional)

Optional Deliverable 2: Project Execution and Outline of Results

- 2.3.15 **Once the programmatic document has been approved, the group should proceed with the research and execution of the course-work**. A high level of autonomy is expected, though groups are encouraged to update the lecturer on significant progress.
- 2.3.16 The group must then research the assigned topic based on the programmatic document.⁹
- ⁹ See [Notes Writing, Ch. 3] for details.
- 2.3.17 The group must identify the central thesis and the web of arguments to support it.¹⁰
- ¹⁰ See [Notes Writing, Ch. 4] for details.
- 2.3.18 Upon completing the research, groups must prepare an outline of results.¹¹
The outline should be detailed and contain all content and arguments, albeit in bullet points or annotated placeholders (e.g., "Here a graph will illustrate...").
- ¹¹ See [Notes Writing, Ch. 5] for details.
- 2.3.19 The outline is not merely a structural guide but should include the complete reasoning and supporting material in the final document.¹²
- ¹² See [Notes Writing, Ch. 5] for details.
- 2.3.20 The outline must be submitted through Microsoft Teams for review. Groups must wait for feedback before proceeding to the final write-up.
- 2.3.21 This is the only stage at which lecturers will comment on the project's results. **However, submission of Deliverable 2 is optional: the group can decide to submit the final output autonomously.**
- 2.3.22 If revisions are requested, a full changelog must be provided. Each group will be allowed at **most one revision** (excluding the original submission). Following the revision, the group should **proceed to the production of the final deliverable**, taking into account

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JUSTIFY WHAT APPROVES WHENEVER WHEN

the feedback provided by the lecturer.

Deliverable 3: Final Output

- 2.3.23 The final deliverable **must be a polished report**.¹³
- 2.3.24 The group must choose a suitable, specific title for their report, which adequately reflects the final content.
- 2.3.25 In particular, the group must:
- Expand the outline into a full draft, eliminating bullet points and placeholders;¹⁴
 - Refine the draft for clarity, coherence, and readability.¹⁵
- 2.3.26 **Revisions will not be requested nor allowed**; once submitted, the deliverable is final.

Deliverable 3: The Final Submission

¹³ See [Notes Writing, Ch. 6 and 7] for details and guidelines.

¹⁴ See [Notes Writing, Ch. 6] for details.

¹⁵ See [Notes Writing, Ch. 7] for details.

4 Use of Generative AI

- 2.4.1 Use of generative AI tools **is permitted subject to transparency and accountability**. In line with current international editorial guidance, any substantive AI-assisted contribution to the work **must be disclosed in the acknowledgements or in a dedicated transparency statement**. This includes substantive assistance with text, code, figures, images, analysis, data processing, or interpretation. The AI system used must be identified, together with the relevant sections and a brief description of the extent of its contribution.
- 2.4.2 **Routine spelling, grammar, and formatting assistance need not be disclosed, although disclosure is encouraged.**
- 2.4.3 **AI systems cannot be listed as authors**. Authors remain fully responsible for the content of their work.
- 2.4.4 If no AI-generated content has been used, disclosure is not required.

DON'T ACCOUNT
IN WORD COUNT

5 Marking Criteria

- 2.5.1 The report is assessed primarily on the intellectual and technical quality of the work undertaken. The central question is not merely whether the report is well written, but whether the group has addressed the coursework aim in a rigorous, well-justified, and substantively meaningful way. A strong submission demonstrates clear problem framing, appropriate and well-executed methodology, convincing evaluation, and careful interpretation of results and implications.
- 2.5.2 **Scope and Problem Framing** assesses whether the group has correctly identified, bounded, and motivated a worthwhile prob-

lem within the coursework brief. It considers the clarity of the research and engineering objective, the appropriateness of the chosen scope, the explicitness of assumptions, and the extent to which the work engages with the real technical or organisational issues raised by the topic rather than retreating into a narrower but easier exercise.

- 2.5.3 **Methodological and Technical Quality** concerns the soundness of the approach taken and the quality of its execution. It evaluates whether methods, datasets, tools, models, baselines, system designs, or analytical frameworks have been selected appropriately, justified properly, and applied competently. Strong work demonstrates methodological discipline, technical correctness, and awareness of alternative approaches.
- 2.5.4 **Evaluation and Evidential Support** assesses the strength of the evidence used to support claims. It considers the quality of experimental design, validation strategy, comparison against baselines or alternatives, treatment of uncertainty and limitations, and the extent to which conclusions are grounded in results rather than assertion. Where appropriate, reproducibility and transparency of the evaluation form part of this criterion.
- 2.5.5 **Interpretation, Insight, and Implications** assesses the depth of reasoning shown in analysing results and drawing conclusions. It includes the ability to explain why results matter, identify limitations and trade-offs, connect findings back to the original problem, and reflect on wider technical, organisational, or societal implications where required by the coursework brief. Strong work demonstrates judgement, not merely description.
- 2.5.6 **Communication and Professional Conduct** assesses whether the work is presented clearly enough to be understood and evaluated, and whether the group has conducted the project responsibly. This includes clarity of structure, intelligibility of figures and tables, consistency across deliverables, adherence to requirements and deadlines, and professional communication during the project. This criterion supports the assessment of the work but is secondary to its substantive quality.
- 2.5.7 Section 5 details the marking criteria and associated mark ranges. The four substantive criteria — Scope and Problem Framing, Methodological and Technical Quality, Evaluation and Evidential Support, and Interpretation, Insight, and Implications — should normally account for the large majority of the mark. Communication and Professional Conduct should play a smaller supporting role. High marks require strong substantive performance; polished presentation or smooth project management cannot compensate for weak problem selection, poor methodology, inadequate evidence, or superficial conclusions.

	Average [18–24]	Good [25–27]	Excellent [28–30L]
Scope and Problem Framing	The report identifies a relevant problem and a workable scope, with a generally clear objective and reasonable assumptions.	The report frames the problem clearly and appropriately, with well-justified scope, explicit assumptions, and a strong alignment between the chosen task and the coursework aim.	The report demonstrates sharp and well-motivated problem framing, with excellent judgement in scope selection and a clear understanding of the deeper technical or organisational significance of the task.
Methodological and Technical Quality	The chosen methods, tools, datasets, or technical approach are broadly appropriate and are applied correctly, though justification or execution may be limited in places.	The work shows strong methodological judgement, technically sound execution, and clear justification of key choices, including awareness of plausible alternatives.	The work demonstrates rigorous and mature methodological design, high technical quality, and compelling justification of decisions in relation to both theory and practice.
Evaluation and Evidential Support	The report presents results and some supporting evidence, but evaluation may be limited in depth, breadth, or rigour.	The report includes meaningful evaluation, appropriate comparisons or baselines, and conclusions that are generally well supported by the evidence presented.	The report provides rigorous and convincing evaluation, with strong validation, careful treatment of limitations and uncertainty, and conclusions that are consistently and transparently supported by evidence.
Interpretation, Insight, and Implications 80/100 mark	The report offers a reasonable interpretation of findings and identifies some limitations or implications, though discussion may remain fairly local or descriptive.	The report analyses findings thoughtfully, discusses trade-offs and limitations clearly, and connects the results back to the wider problem context.	The report demonstrates deep insight in interpreting results, articulating limitations, and drawing broader technical, organisational, or societal implications where relevant to the coursework.
Communication and Professional Conduct	The report is understandable and sufficiently organised for assessment. Requirements are broadly followed and project conduct is adequate.	The report is clear, coherent, and easy to evaluate. Requirements are met consistently, and communication throughout the project is effective.	The report is precise, well-structured, and highly readable. Project conduct is consistently professional, with strong coherence and reliability across all deliverables.

Table 2.4: Marking criteria.

FEDERICO CERUTTI

TIMOTHY J. NORMAN

NOTES ON RESEARCH AND WRITING

0.6 (2025-03-04) — Refinement of content with Adam's checklist for tables and images and how to read a paper.

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1

Introduction

→ I NEED A GROUP (MAIN GOAL)

Writing comprises the following activities:¹

¹ The Tongue and Quill Team 2015.

1. Analyse your purpose and audience to understand with whom you are communicating, what you are trying to convey and why they should care.
2. Research your topic using rigorous scientific methods. (Not generic words, UNRAID) MEANS OF PROBABLY DEFINITION!
3. Support your ideas with well-formed and well-organised argument.
4. Organise and outline your document.
5. Draft.
6. Edit.

Writing is a non-linear activity. While we will discuss them in sequence, several iterations within the various activities are likely to be necessary. Being aware of the entire process is, however, necessary to focus on the goal of producing an effective written document.

2

Analyse your Purpose and Audience (DEL 1)

First, we need to have coherent answers to three questions:

- What do we want to research/write about?
- Why does it matter to our intended audience?
- How do we plan to persuade our audience?

It is paramount to have an answer to these three questions **before** starting to analyse any topic. To this end, a scoping review is needed, see § 3.4.

2.1 What do we want to research/write about?

We live in a world of limited resources. We do not have infinite time and money to address any question in mind. We must, therefore, specify the boundaries of our investigation, and identify the focal point of our writing.

2.2 Why does it matter to our intended audience?

We write to change the way our readers think; we want to have an impact on somebody. They will react only if they see value in what is written. We must keep in mind the intended audience, including their knowledge of the subject, and the intended change we want to trigger in such an audience.

Originality is clearly a fundamental element here: re-writing what somebody else has written has no value.

2.3 How do we plan to persuade our audience?

Once it is clear what our research/writing topic is (and what it's not), we must decide how we want to persuade our audience that there is value in what we write.

How we achieve this depends on the answers to the 'What?' and to the 'Why?' questions. It might be an experimental analysis, a critical literature review or a design document—the 'What?'. We may intend to disseminate key findings from research, argue for novelty of a specific research question

or facilitate the manufacture of a new device—the 'Why?'. Achieving our aims also requires us to define which resources to consider, including the length of the writing piece and the sources of information we build upon.

3

Research your Topic (DEL 2)

We must expand further the rough answers to our 'What?', 'Why?' and 'How?' questions using scientific methods¹ which relies on four pillars:

¹ Gauch 2010.

- Rationality;
- Truth;
- Objectivity; and
- Realism.

RATIONALITY: Rational methods of inquiry use reason and evidence to achieve substantial and specified success in finding truth, and rational actions use reason and justified beliefs to guide choice.

(EX. SO DO INTERVIEW PER X
DELO DIMOSTRARE, DA GRAFIC
? → +)

TRUTH: Investigations must be grounded on measurable objects or events of an external (to us) world, whether this be the natural world or a synthetic one. (INTERVIEWS, DOCUMENTATION, GRAPHS)

→ AND WHAT NOT FROM!

OBJECTIVITY: Objective beliefs concern objects and events of the external world. They can be tested and verified so that consensus will emerge among knowledgeable persons, and they do not depend on controversial presuppositions or special worldviews.

REALISM: Realism is the correspondence of human thought with an external and independent reality: the external world.

3.1 The research topic as a falsifiable hypothesis

We must treat our topic as an hypothesis that must be proved or disproved. We must not limit our investigation to the evidence that supports our hypothesis or idea. We should derive tests that can falsify our hypothesis or idea; that is one of the main characteristics of scientifically testable hypotheses.²

² Popper 1989.

He who knows only his own side of the case, knows little of that. His reasons may be good, and no one may have been able to refute them. But if he is

equally unable to refute the reasons on the opposite side; if he does not so much as know what they are, he has no ground for preferring either opinion. The rational position for him would be suspension of judgment, and unless he contents himself with that, he is either led by authority, or adopts, like the generality of the world, the side to which he feels most inclination. Nor is it enough that he should hear the arguments of adversaries from his own teachers, presented as they state them, and accompanied by what they offer as refutations. That is not the way to do justice to the arguments, or bring them into real contact with his own mind. He must be able to hear them from persons who actually believe them; who defend them in earnest, and do their very utmost for them. He must know them in their most plausible and persuasive form; he must feel the whole force of the difficulty which the true view of the subject has to encounter and dispose of; else he will never really possess himself of the portion of truth which meets and removes that difficulty. Ninety-nine in a hundred of what are called educated men are in this condition; even of those who can argue fluently for their opinions.³

³ John Stuart Mill 1859, p. 67.

Conclusions about hypotheses emerge from logical coordination of evidence and presuppositions.⁴

⁴ Gauch 2010.

PRESUPPOSITIONS are beliefs that are absolutely necessary in order for any of the hypotheses under consideration to be meaningful and true but that are completely non-differential regarding the credibility of the individual hypotheses. Science requires several common-sense presuppositions, including that the external world exists and that we can measure some of its components. These presuppositions also serve to exclude wild ideas from inclusion among the sensible hypotheses under serious consideration.

EVIDENCE is data that bear differentially on the credibility of the hypotheses under consideration. Evidence must be admissible, being meaningful in view of the available presuppositions, and it must also be relevant to the hypotheses.

LOGIC combines the presuppositional and evidential premises, using valid reasoning, to reach a conclusion. Science uses deductive and inductive logic.

IRRELEVANT MATERIAL (i.e., all beliefs that are irrelevant for a given inquiry) must be ignored to avoid infinite and impossible mental processing.

3.2 Parsimony

⁵ Gauch 2010.

Parsimony is the guiding principle when assessing evidence.⁵ The principle of parsimony recommends that from among theories fitting the data equally well, scientists choose the simplest theory. It is also called the principle of simplicity, the principle of economy, and Ockham's razor.

When analysing data always assume that it is the result of data plus noise. Signal ordinarily is relatively simple, caused mostly by only a few major treatment differences or causal factors, whereas the noise typically

is extremely complex, caused by numerous small uncontrolled factors.

Because the signal is parsimonious, it can be captured or fitted readily by an appropriate parsimonious model, but complex noise inevitably requires a complex model

A *model family* is a sequence of models of the same mathematical form that include more and more parameters, such as polynomials vs. linear regressors. Within each family, always consider the complexity of adding more parameters and balance against its role in fitting the signal but not the noise.

Given a model family applied to noisy data, some *statistical criterion* must be specified to determine the best choice. Using cross-validation, for example, to determine the model with lower loss Bayesian Information Criterion (BIC), also called the Schwarz Bayesian Criterion (SBC), which aims at approximating the logarithm of the Bayes factor $\frac{P(E|H_1)}{P(E|H_2)}$ when considering competing hypotheses H_1 and H_2 which both could have generated some evidence.

We must identify the simplest model which gives us good prediction in the general population, without overfitting to the actual data. Going beyond that point (the *Ockham's hill*) brings overfitting while reducing generality to the model.

Scientific experiments generally have two *designs*: a treatment design and an experimental design. The treatment design specifies the deliberately controlled factors of scientific interest. The experimental design specifies how the treatments are allocated to the experimental units, which usually involves randomisation and replication to reduce bias and increase accuracy.

3.3 Write notes!

At this stage we write notes, which must be clear and focused.

CLEAR: we will have to elaborate further those notes, hence they cannot be just broken sentences or technical results from an experiment, rather fully developed sentences with context and references.⁶

Ethics, copyright laws, and courtesy to readers require authors to identify the sources of direct quotations or paraphrases and of any facts or opinions not generally known or easily checked. [...] Source citations must always provide sufficient information either to lead readers directly to the sources consulted or, for materials that may not be readily available, to enable readers to positively identify them.⁷

⁶ It is highly recommended to use a tool such as Zotero for collecting, organising, and take notes of your references.

⁷ The University of Chicago Press Editorial Staff 2017, § 14.1.

FOCUSED: deciding in advance the answer to the 'What?' question helps us here in deciding what is out-of-topic, thus forcing us to stay on message. We must read literature and run experiments with the intent of further understanding our research topic. We cannot just report what we observe or what we read in a source: we must be able to explain it to our readers.

⁸ B. Franklin and W. T. Franklin 1818, p. 3.

I would advise you to read with a pen in your hand, and enter in a little book short hints of what you find that is curious, or that may be useful; for this will be the best method of imprinting such particulars in your memory, where they will be ready, either for practice on some future occasion, if they are matters of utility; or at least to adorn and improve your conversation, if they are rather points of curiosity.⁸

While this suggestion clearly applies to reading, it also applies to any other experience, from pure observation, to documenting software design.

3.4 Literature review

⁹ Jesson, Matheson, and Lacey 2011.

There are two types of literature review: traditional and systematic.⁹

A TRADITIONAL LITERATURE REVIEW is a written appraisal of existing work on a topic, with a personal interpretation, and a critical assessment examining methods and results.

¹⁰ Jesson, Matheson, and Lacey 2011.

Literature reviews can serve specific purposes:¹⁰

Conceptual review: contributing to a better understanding of the issues;

State-of-the-art review: bringing readers up to date on the most recent research on the subject;

Expert review: expressing the point of view of an acknowledged expert, which might be heavily influenced by the writer's personal selection of material; or

Scoping review: setting the scene for a future research agenda, documenting what is already known and the gaps. It is also used as the first step in refining the questions for a subsequent systematic review.¹¹

¹¹ To answer the 'What?', 'Why?', and 'How?' questions, § 2, a scoping review is often required.

3.5 Selecting your sources

In conducting literature reviews, you are faced with a dizzying array of potential sources. Scientific papers published in peer reviewed journals, conference proceedings, etc. are most common. They do vary in reviewing standards, however, and it is your responsibility to review material critically and select your sources carefully.

When starting from scratch in a new research area, reading a review article can help you more rapidly develop a good picture of relevant literature. These may be out-of-date, however, especially in a fast-moving field of research.

It is worth being as rigorous as you can in identifying the literature you will rely on in writing your review. A good process to use is:

1. Identify a set of keywords or queries (groups of keywords) that reflect your area of research interest. Try to focus on terms that are more unique to your area, but more general terms can help identify highly cited articles in the wider context.

2. Select the repositories you will use to search for relevant literature such as Google Scholar, IEEE or ACM Digital Libraries, or similarly indexed and cross-referenced tools/resources.
3. For those articles you identify as potentially relevant (i.e. after reading the title and abstract only), make a note of things like the venue. There is typically a strong correlation between topic area and venue because researchers who are innovating in your area want others with common interests to read their work. Articles are also typically peer reviewed by those with common expertise in such venues, and hence claims made in the article are more likely to be trustworthy.
4. Often you find useful articles on repositories such as arXiv. These could have been peer reviewed and published elsewhere. Is this the same as that checked by reviewers, an older draft (possibly with errors), a corrected version, an extended version? Help your reader by citing the most trustworthy version, which is typically the one in the peer reviewed conference/journal unless you are relying on extensions.
5. Cluster and filter the articles you've identified. Clusters of articles can help you build a map of techniques or methods used to solve a problem or sub-areas of interest. Filter the articles to identify those you will read more carefully, which could be a few representative (most recent, or most highly cited, etc.) for each cluster.
6. Take a note of the *full bibliographic reference* of the articles selected using a consistent style. Do not rely on automatically generated BibTeX entries; they are not consistent across repositories.
7. Reflect on your keywords and/or repositories and iterate.
8. Once you have collated a good number of high-quality articles using the keywords and repositories selected, then expand out from what you judge to be the most important/relevant articles using citations. This helps to ensure you haven't missed any hidden gems.

Citing articles: one of the most useful functions of systems such as Google Scholar. Follow the "Cited By" link—these authors agree with you that the paper you've read is interesting; they may have built upon the work or criticised it¹².

Cited articles: look up articles cited by the paper, but check their citation profile/longevity.

¹² They could, of course, be the same authors; a self-citation.

3.6 How to read a paper

To form an educated opinion on a scientific subject, reading primary research literature is essential. Scientific papers require a unique reading approach and demand note-taking, multiple readings, and supplementary research. Primary research articles consist of sections like abstract, introduction, methods, results, and conclusions/discussion. Assessing the authors' affiliations and publishing journal's credibility is crucial.

The content of this section is a summary of Dr. Raff's blog post available at <https://rb.gy/8t9t1x> (on 23rd Feb 2024).

The reading process involves several key steps:

- Begin with the introduction rather than the abstract to avoid bias.
- Identify the overarching question of the field to contextualise the research.
- Summarise the background in five sentences or fewer to grasp the research context.
- Note the specific question(s) the research aims to answer.
- Understand the authors' methodological approach.
- Carefully read the methods section, using diagrams if necessary for clarity.
- Summarise the results for each experiment, figure, and table, reserving interpretation for later.
- Assess if the results address the research question(s) and consider your interpretation.
- Review the conclusion/discussion sections to compare your interpretations with those of the authors.
- Revisit the abstract to evaluate its consistency with the paper's content and your understanding.
- Lastly, explore other researchers' perspectives on the paper for broader insights.

3.7 Critiquing the literature

When reading a paper, it is helpful to keep in mind a number of questions while critically assessing a source.

General: what to consider about the paper as a whole.

- What is the purpose/aim of the paper, and how might this influence the way it is written, the methods used and its contribution?
- Are their research decisions well justified, and are they reasonable?

Assumptions: the starting point of the research.

- What assumptions are being made?
- Are these assumptions reasonable in this context?
- Are assumptions grounded on reliable/authoritative evidence?
- Are there unstated assumptions that would affect the authors' arguments?
- Are concepts being conflated?
- Is anything being used out of context?

Process: the research methods used in the paper.

- What kind of data is being used, and is this a suitable choice?
- Are there structural biases in the data being employed?
- If an existing model is being used, is it being employed in a reliable way? (E.g. consider inherent biases in models such as GPT.)
- Is the experimental data presented clearly?
- Do experiments sufficiently cover all important cases?
- Are appropriate methods used for data analysis, and are they applied rigorously?
- Are appropriate proof methods used for key claims?
- What is missing from the paper?

Claims: the truth claims being made in the paper.

- Are the claims made more general than the evidence warrants (hasty generalisation)?
- Are the claims too certain, or are other false arguments being employed (see § 4.2)?
- Are the conclusions supported from the analysis of the results?
- Is the data available, and are results reproducible?

4

Support your Ideas (DEZ 2)

It is now the time to start organising our notes in a coherent map to achieve the desired impact on our audience. We must determine the strategy we want to follow to persuade our reader. Think about a web of arguments in favour of the claims that are the most convincing for the reader, while actively undermining possible counterarguments evident in prior research.

4.1 Logical arguments

The fundamental elements of a logical argument¹ are:

- Claim or Conclusion;
- Evidence that supports the claim; and
- Warrants linking evidence to the claim.

¹ The Tongue and Quill Team 2015; Booth et al. 2003.

THE CLAIM OR CONCLUSION is our position on an issue, our answer to a controversial question, or our recommendation for resolving a problem.

THE EVIDENCE that supports the claim can be of different types. It can be:

- Definition: a precise meaning or significance of a word or phrase
- Testimony: comments of recognised authorities to support your claim.
These comments can be direct quotations or paraphrases. Citation must be clear and complete.
- Statistics: a summary of data that allows the audience to better interpret quantitative information.
- Explanation: makes a point plain or understandable or gives the cause of some effect. Possible explanatory techniques are:
 - Analysis: clarifying a complex issue by examining one piece at a time;
 - Comparison and Contrast: comparison to dramatise similarities between a number of objects or situations and contrast to emphasise differences; and

- Description: a detailed account with enough information for a willing reader to assess the quality of the explanation.

A good supporting piece of evidence must:

- come from a trustworthy source;
- be accurate;
- be precise;
- be relevant; and
- be sufficient for providing explanation.

WARRANTS THAT LINK EVIDENCE TO THE CLAIM. There are often stated or unstated assumptions that link the evidence to the claim and explain why the evidence is relevant to the argument. We also have to state the qualifications that would limit the claim; i.e. the limits of our arguments.

4.2 What to avoid: Flawed arguments

² The Tongue and Quill Team 2015.

A partial list of flawed arguments to be avoided at all costs:²

Asserting the conclusion The practice of slipping in an assertion and passing it off as a fact. Circular reasoning (also known as begging the question) involves rewording your claim and trying to use it as evidence.³

³ E.g., A is true because B is true; B is true because A is true.

⁴ E.g., "Do you think John Smith is going to improve his drinking problem?"

Loaded question A question with an assertion embedded in it.⁴

Ad Hominem Involves an assault on your opponent as an individual, instead of their position.

Emotional appeals These might be:

- to force, targeting the audience's fear;
- to pity, targeting the audience's compassion;
- to popularity or tradition, such as stirring symbols.⁵

⁵ E.g., "good management principles demand we take this action".

False authority Accepting facts based on the opinion of an unqualified authority

False cause (Post Hoc) When you assume one event causes a second event merely because it precedes the second event.⁶

⁶ E.g., many people observe that Event B occurred after Event A and conclude that A caused B. This is not necessarily true; maybe a third factor, Event C, caused both A and B.

Single cause When you assume there is a single cause for an outcome, when in fact multiple causes exist.

Faulty analogy When irrelevant comparisons are put forward.

Faulty dilemma When only two options are entertained, with no middle ground.

Hasty generalisation Jumping to a conclusion with insufficient evidence.

Non sequitur The conclusion does not necessarily follow from the facts presented.

Slippery slope If we take one small step in an unpleasant or dangerous direction, we'll have to go all the way; like slipping down a hill.

Red Herring Bringing up irrelevant information to get the audience off track.

Stacked evidence Withholding facts or manipulating support so that the evidence points in only one direction.

Straw man Attacking a weaker, grossly simplified version of the opponent's argument rather than directly addressing the argument presented.

Organise and Outline (DEZ 2)

After having devised the map of arguments and counterarguments that constitute our communication strategy for ensuring that our reader sees value in what we write, we must **design the delivery of the document**.

Most technical and scientific documents follow a structure like this:

1. Introduction

It should contain the motivation, hence answering the 'Why?' question, and a brief summary of the rest of the document. **It must be the last section you write** and it should be accessible to a broad audience, probably wider than your intended audience. Try to avoid unnecessary technical details. Even more importantly, try to avoid formal statements such as equations or overly complicated diagrams. Instead, it is much better to try to have an intuitive example that motivates the document.

2. Background

It must contain a brief account of where you started your investigation and a summary of the knowledge a reader must have to appreciate your work. The argument for your work being novel may be supported by a scoping review (see § 3.4). A scoping review will be more detailed in a thesis/dissertation. For a conference/journal paper the key challenge here is what is required in order for the reader to understand the technical content of the document.¹

3. Methodology

The main goal of the document (in brief) and how do we want to achieve it. It is very important to list what we will not discuss in the document.

4. Results

Critical assessment of the achieved goal.

5. Discussion

An exposition of how your contributions fit in the wider literature. A discussion of the implications of your work and its applications. Reflection and honest critical review of your work.

¹ Keep this brief for a paper or integrate into the introduction and consider balance between background and discussion; follow the excellent guidance from Simon Peyton-Jones <https://simon.peytonjones.org/great-research-paper/> (last accessed 01/07/2022)

6. Conclusions

Summary of the document, which may include a discussion of future work. **It must be written just before writing the Introduction**

The **outline** of your document should be a list of bullet points for each of the sections comprising the structure of the final writing. This list must represent a **linear path in the map of arguments and counterarguments** developed in the previous step to illustrate how they will be expanded in the text.

Reflect on your planned structure:

- Within any enumerated section there may be two or more subsections, **never one**. For example, **there should be no Section 2.1 unless there is also a Section 2.2**.
- If considering an additional series of subsections, ask yourself whether this is necessary. For example, if you expect to have only one or two paragraphs in a section, you have imposed too much formal structure and this will affect the flow of your narrative.

6

Draft

(DEZ 3)

Paragraphs are the fundamental units in writing. A paragraph is a group of related sentences that discuss one, and usually only one, main idea. The paragraph should be as long as it is strictly necessary to develop the main idea clearly.

6.1 Paragraphs

Each paragraph comprises three parts:

- Topic sentence;
- Supporting sentences;
- (Optional) Concluding sentence.

TOPIC SENTENCE: states the main idea of the paragraph, considering both the main topic and what of this is addressed in the paragraph. It is the most general statement in the paragraph because it gives only the main idea, and no specific details. In assert-justify, this is the first sentence in the paragraph. In general, it should stand-out: if, as usual, it's at the beginning of a paragraph, it must be indented and/or separated from the previous paragraph.

THE FIRST LINE
IS THE MOST IMPORTANT!

PROBABLY SUB
IT FIRST (AND
ONLY?)

SUPPORTING SENTENCES, each of which must directly explain or prove the main idea of the paragraph (*coherence*). Typically one paragraph is a single argument or a few chained arguments together supporting the main idea (or conclusion of the argument). Figures and tables can be referenced in the supporting sentences, adding a few words summarising them and telling the reader the main message underpinning each one.

CONCLUDING SENTENCE: (optional) leaves the reader with important points to remember. Concluding sentences are customary for standalone paragraphs. Paragraphs that are parts of a longer piece of writing usually do not need concluding sentences

6.2 Sentence realisation

¹ The Tongue and Quill Team 2015.

Paragraphs are composed by sentences, which must be realised by keeping in mind the principles of effective communication.¹ Communication must be:

- focused;
- organised;
- clear;
- understanding; and
- supported.

FOCUSED. We must address the Issue, the whole issue and nothing but the issue. Failure to focus comes in three forms: answering the wrong question; and/or answering only part of the question; and/or adding irrelevant information.

ORGANISED. Systematically present information and ideas. A variety of common paragraph patterns can be employed.

Topical or Classification Pattern: present groups of ideas, objects or events by categories. For instance, present general statements followed by numbered listings of subtopics to support, explain or expand the statements.

Comparison and Contrast Pattern: discuss similarities and/or differences between topics, concepts or ideas.

Chronological Pattern: discuss events, problems or processes in the sequence of time in which they take place or should take place.

Sequential Pattern: similar to chronological pattern, but usually here the timing is not as important as the specific order.

Spatial or Geographical Pattern: similar to the chronological pattern, but considering 2, 3, or 4 dimensions altogether.

Problem and Solution Pattern: discuss all facets of the problem: its origin, characteristics and impact. When describing the proposed solution, include enough support to convince your readers the solution is practical and cost effective.

- One Solution: discuss the problem and follow with the single, most logical solution.
- Multiple Solutions: discuss the problem, several possible solutions, the effects of each and your recommendation.
- Multiple Solutions, Pro-Con: this popular format includes a discussion of the advantages ("Pros") and disadvantages ("Cons") of each solution.

Cause and Effect: show how one or more ideas, actions or conditions lead to other ideas, actions or conditions. Two variations of this pattern are

possible: begin with the effect, then identify the causes; or begin with the causes, then identify the effects.

Reasoning or Logic Pattern: see § 4.

CLEAR: communicate with clarity and make each word count. Get to the point, not hide ideas in a jungle of words. Master the rules of the language, including how to spell and pronounce words and how to assemble and punctuate sentences.

Buy, read and follow a guide to grammar and punctuation. This will help you develop a clear, consistent and correct style of presentation. Tools such as Grammarly can help, but do both and try to learn from tools like Grammarly rather than just use them as a crutch.

Punctuation that is often confused:

The colon: Used as a pointer from one statement to something more specific. *The scientific method relies on four pillars: rationality, truth, objectivity and realism.*

The semi-colon: Used to separate two complete, but closely related sentences; often one that includes an example related to the other. Also used to separate items in a bulleted or enumerated list as in the examples above.

The hyphen: In $\text{\LaTeX}$ -. Used to connect words to make a compound word or to avoid ambiguity. Never add spaces either side of the hyphen. *Man eating shark vs. man-eating shark*

The en dash: In $\text{\LaTeX}$ --. Used for a range of numbers. Never add spaces either side of the en dash. *0–1.*

The em dash: In $\text{\LaTeX}$ ---. Used as an alternative to brackets for a hard interruption. Guides differ, but spaces either side of the em dash make the presentation less cramped and ugly. Most punctuation guides prefer brackets or bracketing commas.

The apostrophe: indicates missing letters. Be careful that it's means "it is" whereas its is the possessive form, "it has".

The comma: multiple uses. Can be used to "bracket" an aside in a sentence; see final sentence in the second paragraph of § 2.3. Can be used to delimit items in a short list. Can be used to indicate an aside at the beginning and end of a sentence, *for example*.

Quotation marks: In $\text{\LaTeX}$ be careful *not* to use "; use ' and ', and ' and ' ' to quote 'one thing' or "another".

UNDERSTANDING: understand your audience and its expectations, see § 2.

SUPPORTED: see § 3 and 4.

6.3 Checklist

- ☐ Use short sentences with few subordinates.
- ☐ Clarify who/what is the subject of each sentence: that must be the main focus, and that determines whether to use an active or a passive form. As a preference, use active over passive form.
- ☐ Before using a technical term, define it.
- ☐ Before using an abbreviation or an acronym (an abbreviation that is pronounced as a word), define it. Moreover, use these sparingly and only if the extended version would be too long.
- ☐ When citing a reference, add at least a few words/one sentence illustrating its main message. If you cite something, you are expected to have read it.
- ☐ Use synonyms sparingly and only if there is not a specific technical term. In technical writing, repetition is good.

7

Edit

(DEL 3)

After drafting the document, we now need to proceed to polish it to ensure it is understandable and will be appreciated by the intended audience. *Easy reading requires hard writing.*¹

¹ <https://quoteinvestigator.com/2014/11/05/hard-writing/> (last accessed 21/06/2022)

7.1 The Big Picture checklist

- ☐ Can a reader answer the questions 'What?', 'Why?', and 'How?' after reading the text? *CONSISTENT WITH DEL 2 (IF NOT, WHY?)*
- ☐ Does the answer feature from the introduction, albeit in a short format?
- ☐ Do the introduction and the conclusion sound like they go together without being identical? Does the introduction declare the purpose and does the conclusion show the readers that we accomplished our purpose?
- ☐ Does each chapter provide the readers with a short introduction and a short conclusion, to help them navigating the text?
- ☐ Check for relevance and completeness, against the map of arguments and the outline designed earlier:
 - Are some paragraphs irrelevant or unnecessary?
 - Is there any major points missing?
 - Are paragraphs arranged in a consistent order?

7.2 The Paragraph Structure and Clarity checklist

UNITY OF FOCUS

- ☐ Is there one, and only one, main point of the paragraph?
- ☐ Is all the information in the paragraph related enough to be in the same paragraph?
- ☐ Can the central idea of each paragraph easily identifiable?

TOPIC SENTENCE

- ☐ Does the paragraph have a topic sentence; i.e. one sentence that captures the central idea of the paragraph?
- ☐ Is the topic sentence the first sentence of the paragraph? (Or, when starting with a transitional sentence, the second sentence?)

SUPPORTING IDEAS

- ☐ Do sentences expand, clarify, illustrate and explain points mentioned or suggested in each main idea? The goal is to smoothly lead the reader, step-by-step, to each main idea.
- ☐ Are there enough details in the paragraph to support the central idea?
- ☐ Are there any “extra sentences” that seem to be irrelevant to the main point?
- ☐ Are there any statements that are vague or lacking justification/evidence?
- ☐ Do all transitional words, phrases, and clauses improve the flow and show proper relationships?
- ☐ Are all the citations properly referenced? You can use the Chicago Manual of Style² or similar. Also make sure to follow the convention discussed in the following, Section 7.3

² The University of Chicago Press Editorial Staff 2017.

7.3 Referencing and Citation Management

When writing your report, you must reference each document you cite correctly. To ensure accuracy and consistency in your bibliography, check whether the document is available through Google Scholar.³

If the document is found, use the “Cite” function, which provides formatted citations in various styles. Many references also have direct export options for citation management software, such as BibTeX, which is particularly useful for L^AT_EX users. If you are using Microsoft Word, citation files can be imported into reference managers such as Zotero, Mendeley, or EndNote.

To streamline the citation process, it is strongly recommended to use a reference management tool such as Zotero,⁴ which integrates with both L^AT_EX and Word. Zotero allows you to store references, generate bibliographies, and insert citations directly into your document. When using L^AT_EX, you can export citations in BibTeX format and include them in your .bib file.

If the document you need is unavailable through Google Scholar, take the following steps to ensure proper referencing. Create a full bibliographic entry if the document has identifiable metadata such as a title, author, publication year and an associated institution (e.g., Bank of England). If the document is accessible online, provide its URL along with the last access date to indicate when you retrieved the source. This is particularly

³ <https://scholar.google.com/> (on 4 March 2025). Google Scholar indexes various academic sources, including journal articles, conference papers, books, and technical reports.

⁴ <https://www.zotero.org/> (on 4 March 2025).

important for online reports, working papers, or technical documents that may not be permanently archived.

For less formal sources, such as *blog posts, discussion forums, or other non-academic materials*, evaluate their relevance to your report before citing them. If you consider them useful, include their URL in a **footnote** rather than in the primary bibliography. Always indicate the **last access date**, as online content can change or become unavailable over time. While informal sources can provide valuable context, they should be used cautiously, ensuring they complement rather than replace peer-reviewed or authoritative references.

7.4 Figures and Tables

- ☐ Do you have permission to reproduce an image? Who created it? Where did you obtain it? What license is the image shared under?
- ☐ For each figure and table, is present a paragraph (at least 50 words) in the text discussing it?
- ☐ Is the figure/table useful? For what purpose? Is it clear the purpose to the reader?

7.5 The Sentence, Phrases, and Word checklist

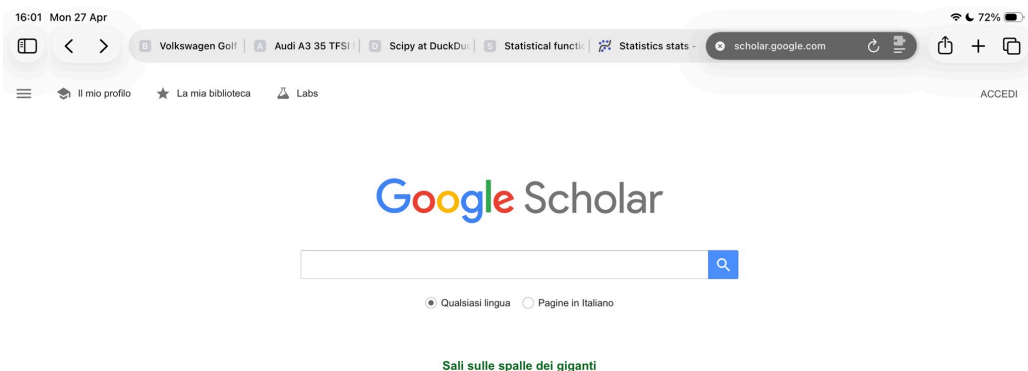
- ☐ Check the grammar, for instance using Grammarly (in English)⁵ or Language Tool (multi-language).⁶ Use pandoc⁷ for transforming the manuscript in different formats if needed
- ☐ Read the document aloud or have a speech synthesis system doing that (Microsoft Word for instance has one integrated). This will help you to check whether the sentences flow as expected and whether there are semantic mistakes the automatic grammar checker failed to identify.
- ☐ Speed-read the document by just reading the first sentence of each paragraph (or at most the first two); can a reader get a grasp of the main message?

} SEMANTIC
REVIEWED
DATA GRAPHS

⁵ <https://www.grammarly.com/> (on 10th June 2022)

⁶ <https://languagetool.org/> (on 10th June 2022)

⁷ <https://pandoc.org/> (on 10th June 2022)



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